

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Semester : VI

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/

Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: BTEEC602_Y22 Electrical Machine Design

Max Marks: 60

Date: 18/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Provide a brief explanation of the study of magnetic, electric, and dielectric materials	CO1	6
B) Explain the key principles of design. How do these principles impact user experience and overall success of a design?	CO3	6
C) State the limitations in the design of Electrical Machines? Explain in brief.	CO4	6
Q.2 Solve Any Two of the following.		12
A) Explain the fundamental differences in the design principles for AC and DC machines. How do these differences impact factors like magnetic circuit, windings, and commutation?	CO2	6
B) What are the primary differences between AC and DC windings in terms of design considerations and their applications in electrical apparatus?	CO2	6
C) Describe the impact of core materials on the design and performance of AC and DC windings in simple electrical apparatus.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) What are the primary design considerations for the stator in an induction motor, and how do they impact motor performance?	CO1	6
B) How does the choice of stator core material influence the efficiency and magnetic performance of an induction motor?	CO4	6
C) Write expression for Calculation of Ampere-Turns for flux distribution in rotating machines.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain Selection of length of air gap and explain factors affecting length of air gap, design of rotor.	CO2	6

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| B) Explain with neat diagram harmonic field effect on the performance of 3-phase induction motor | CO3 | 6 |
| C) Discuss the impact of skewed rotor slots on the operation and performance of squirrel cage rotors in induction motors. | CO1 | 6 |

Q. 5 Solve Any Two of the following. **12**

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| A) Discuss the significance of temperature rise limits in electrical machines and their impact on machine performance and longevity. | CO2 | 6 |
| B) Explain how the cooling time constant is calculated and its importance in predicting the rate of temperature decrease in an electrical machine. | CO4 | 6 |
| C) Discuss the factors that influence the heating time constant and its relevance in different types of electrical machines. | CO3 | 6 |

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